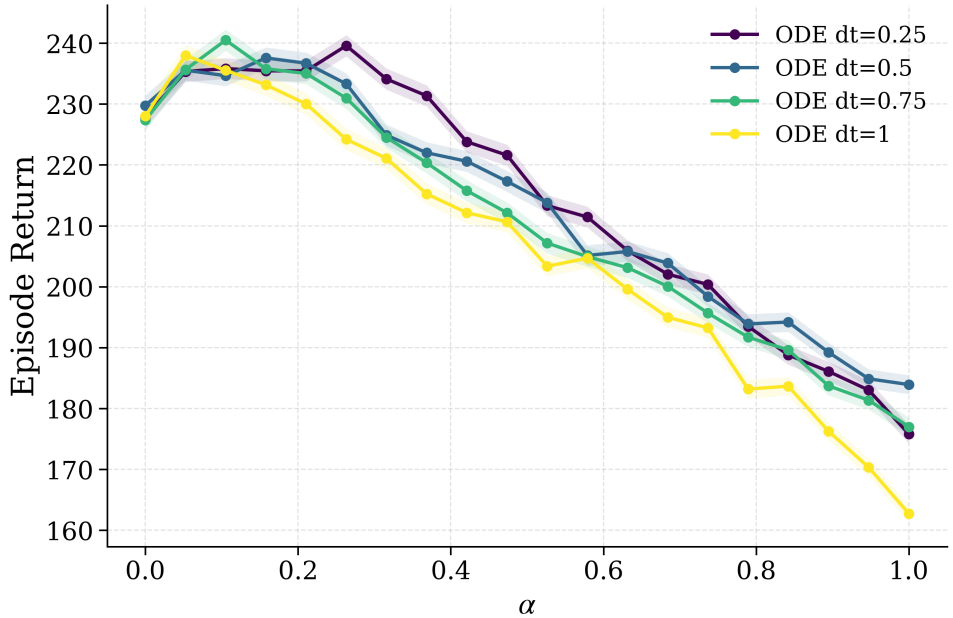
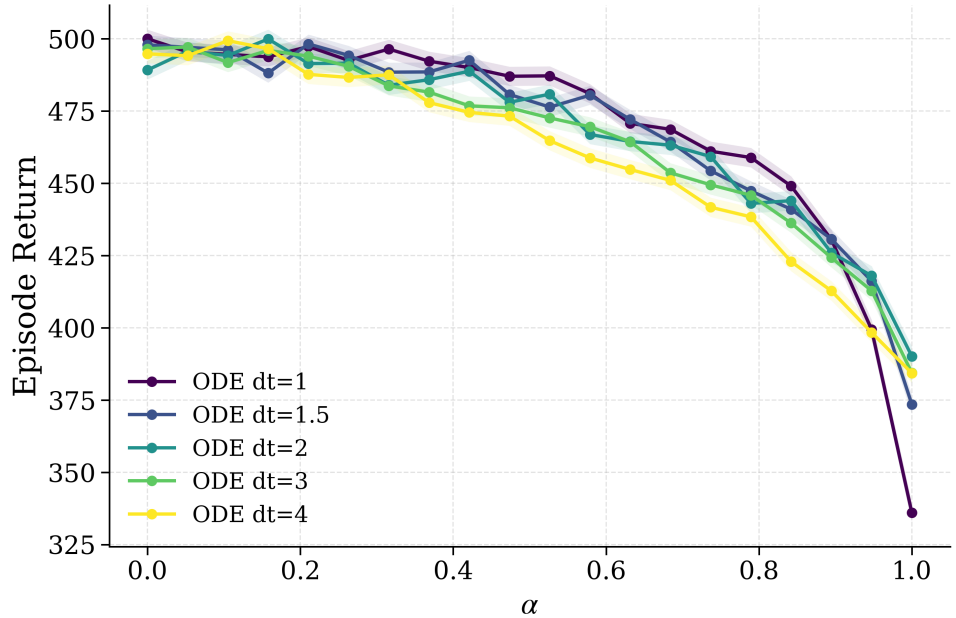
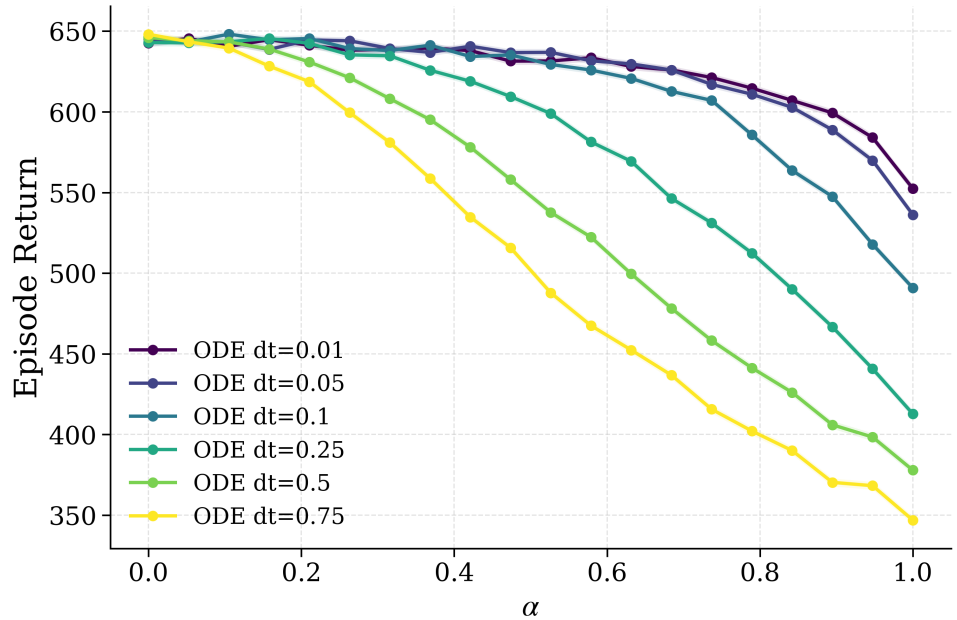


(a) HopperHop



(b) AcrobotSwingup



(c) FishSwim

Figure 1: Q-function guidance in DME-REPPO. The guided control $\tilde{u}_\theta(\tilde{s}_t)$ is defined as a weighted combination of the learned control $u_\theta(\tilde{s}_t)$ and the Q-function gradient: $\tilde{u}_\theta(\tilde{s}_t) = (1-\alpha)u_\theta(\tilde{s}_t) + \alpha\frac{dt}{\mathcal{T}}\nabla_a Q(\tilde{s}_t, a)\big|_{a=a_t^k}$. Here, $\tilde{s}_t = (s_t, a_t^k)$, $\mathcal{T}$ is the final training temperature, α is an interpolation parameter, and dt is the step size, both of which require tuning.